Recursion Problem Bank

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Legend
★ = Core Placement Problem (Mandatory)
Language-Agnostic DSA Problem Bank | Solve before Linked List, Tree, and Graph.
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Section 1: Recursion Mechanics (Internals)

★ **Print Numbers from 1 to N using Recursion**
★ **Print Numbers from N to 1 using Recursion**
★ **Sum of First N Natural Numbers using Recursion**
★ **Factorial of N using Recursion**
Multiply Two Numbers using Recursion (Repeated Addition)
★ **Power of a Number using Recursion (x^n)**
Print Each Digit of a Number using Recursion
Count Digits of a Number using Recursion

Section 2: Head and Tail Recursion

★ **Demonstrate Head Recursion with Print**
★ **Demonstrate Tail Recursion with Print**
Convert Head Recursion to Iterative Loop
Convert Tail Recursion to Iterative Loop
★ **Sum of Array Elements using Head Recursion**
Sum of Array Elements using Tail Recursion

Section 3: Tree Recursion

★ **Fibonacci Number using Recursion**
Count Number of Recursive Calls in Fibonacci
★ **Staircase Problem — Count Ways to Climb N Steps (1 or 2 at a Time)**
Tile Problem — Count Ways to Fill 2xN Board
Count Binary Strings of Length N without Consecutive 1s

Section 4: Recursion with Memoization (Bridge to DP)

★ **Fibonacci using Recursion + Memoization (Top-Down DP)**
★ **Staircase Problem using Recursion + Memoization**
★ **Coin Change — Minimum Coins using Recursion + Memoization**
Longest Common Subsequence using Recursion + Memoization
0/1 Knapsack using Recursion + Memoization
Count Unique Paths in Grid using Recursion + Memoization

Section 5: Recursion on Arrays and Strings

★ **Find Maximum Element in Array using Recursion**

★ **Find Minimum Element in Array using Recursion**
★ **Check if Array is Sorted using Recursion**
★ **Reverse an Array using Recursion**
★ **Check if a String is Palindrome using Recursion**
Binary Search using Recursion
Find All Occurrences of an Element using Recursion
Sum of Digits of a Number using Recursion

Section 6: Backtracking Problems

★ **Generate All Subsets of an Array (Power Set)**
★ **Generate All Permutations of a String**
Generate All Permutations of an Array
★ **N-Queens Problem**
Rat in a Maze Problem
Sudoku Solver
★ **Word Search in a 2D Grid**
Generate All Combinations of Size K from N Elements

Section 7: Divide and Conquer using Recursion

★ **Merge Sort using Recursion**
★ **Quick Sort using Recursion**
★ **Find Maximum Subarray using Divide and Conquer**
Count Inversions in Array using Merge Sort
Exponentiation by Squaring (Fast Power)

Section 8: Advanced Recursion Patterns

★ **Tower of Hanoi**
★ **Josephus Problem**
Generate Balanced Parentheses
★ **Letter Combinations of a Phone Number**
Decode Ways (Number to Letter Mapping)
Print All Paths in a Maze from Top-Left to Bottom-Right
★ **Combination Sum (Pick Elements that Add to Target)**

Section 9: Real-World Applications of Recursion

★ **File System Directory Traversal using Recursion (List All Files)**
★ **Recursive Descent Parsing of Arithmetic Expressions**
JSON / XML Tree Parsing using Recursion
★ **Directory Size Calculator using Recursion**
Fractal Pattern Generation — Sierpinski Triangle
Tower of Hanoi — Trace Disk Movement Step by Step